CHEM 110 Spring 2026 S3

Assignment 2

Due on Monday (January 19th) 6 pm

(To get the full points, you need to write down the solving or calculation process)

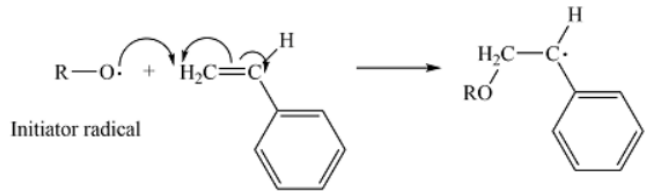
1. Determine the polarity (polar or nonpolar) for the following molecules. Explain your reason by drawing their molecular geometries and dipole moments.

(1) $BrCl_3$

(2) CS_2

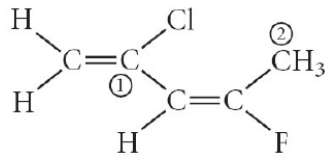
(3) SiF_4

2. Styrene ($C_6H_5CH{=}CH_2$) is the compound that is used to synthesize plastic polystyrene (PS). In the initial step, an initiator radical (R-O·) attacks the double bond in styrene to start the polymerization reaction.



Is tne reactivity for the two bonds in the carbon-carbon double bond ($C_6H_5CH{=}CH_2$) the same when reacting with radicals in the polymerization reaction? Explain your reasoning briefly

3. Determine the electron geometry, molecular geometry, and hybridization type for the two labeled carbon atoms in the following molecule.



4. Write down the BEST Lewis structure for the following molecules and verify why it is best by using formal charges.

(1) SO_3

(2) $COCl_2$

(3) Nitrate anion NO_3^-

5. Determine the electron geometry and molecular geometry of the atom(s) with an underline in the following molecules

(1) $H_2\underline{C}=O$

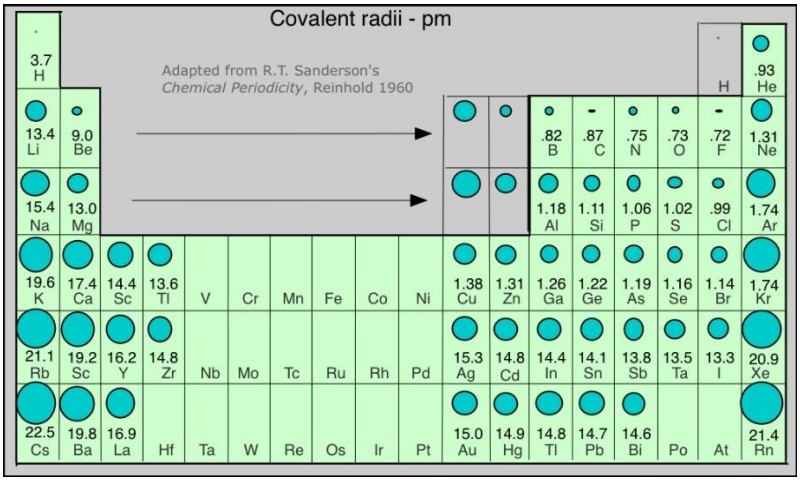
(2) $H_2C=\underline{N}H$

(3) $H_3C-C\equiv\underline{C}-\underline{O}-CH_3$

6. Estimate and rank the bond polarity for the following compounds using electronegativity. Explain your reasoning briefly

CO I_2 RbCl IBr NO

7. In the old times, like the 1960s, people ranked noble gas elements as one of the largest atoms in a period. It contradicts what we observe nowadays using advanced techniques. Explain the reason behind this difference. (Hint: the definition of the “radius” or the way they used to estimate the “radius”)



8. TRUE or False

_____(1) Bond energy is the energy required to break one bond in the gas phase.

_____(2) Based on MO theory, H_2 is less stable than H_2^+.

_____(3) The elements in the same period have similar chemical properties.

_____(4) Lithium should be more reactive than beryllium.

_____(5) According to the Octet rule, a central atom must bond with four other atoms to be stable.

_____(6) CH_2CHCH_3 contains 8 sigma bonds and 1 pi bond.